



Safe writes for AI agents on Exasol.

The agent never touches production. It works on a branch, a person sees the exact rows that would change, and every merge can be undone exactly.

Governed changes

Observed impact

THE PROBLEM

AI agents can now reach your database. Writing is where it gets dangerous.

Two customer databases after an acquisition. An agent is asked to merge them, and a father and son at one address become a single customer: one person's loan history, gone. The SQL looked fine.

READING

Agents connect through MCP and query data. Low risk: nothing changes.

WRITING

One wrong `UPDATE` or a bad merge silently corrupts the customer master.

TODAY

Exasol's official MCP server ships with writes switched off. The safe write path is missing.

Approving an agent's **SQL text** is guesswork: it is a prediction of what the statement will do. Reviewers need to see the **effect**.

THE SOLUTION

Approve the effect, not the SQL.

1

Branch

The agent gets a private copy of the tables it writes, inside Exasol.

2

Observe

Any SQL runs on the branch. Exasol hashes every row to compute the exact diff.

3

Gate

Small, safe changes pass. Risky ones wait for a person, who approves row by row.

4

Merge

Staged, fingerprint-checked, then swapped in atomically.

5

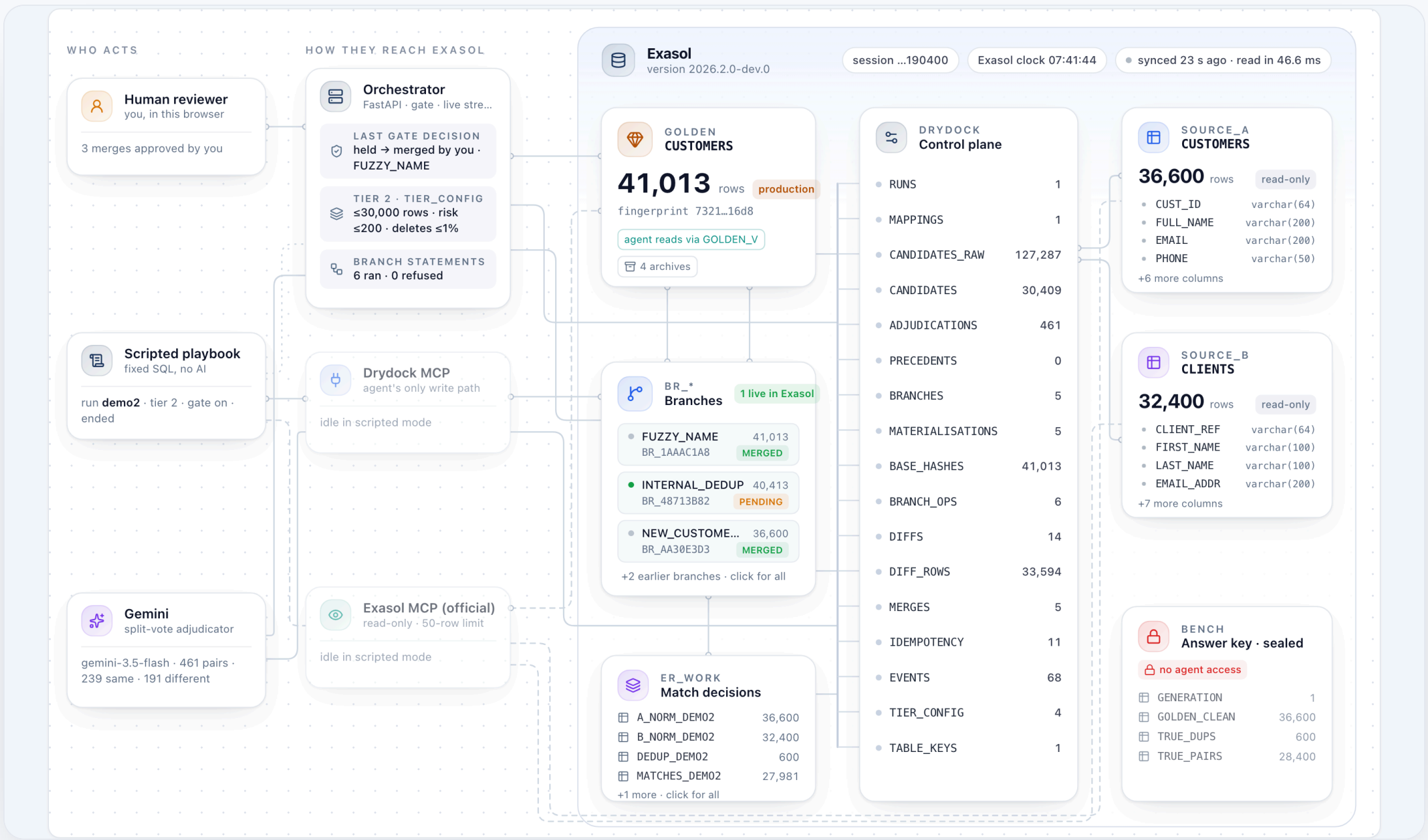
Undo

Unmerge restores the table; the fingerprint proves it byte for byte.

Discarding a branch is `DROP SCHEMA`. Production was never touched, so there is nothing to roll back.

HOW IT WORKS

Every moving part is a real event. Every number is what Exasol reports.



No extra infrastructure. Exasol does the work.

- **Branch:** `CREATE TABLE ... AS SELECT` into its own schema
- **Diff:** `HASH_SHA256` per row, full outer join on the key
- **Match:** `EDIT_DISTANCE` , `SOUNDEX` , date maths, in SQL
- **Swap:** two `RENAME` s in one transaction
- **Discard:** `DROP SCHEMA ... CASCADE`

≈1 s

copy of the 36,600–41,013-row
customer table

≈1 s

row-level diff of 24,186 changed
rows

15–35 ms

the atomic swap into production

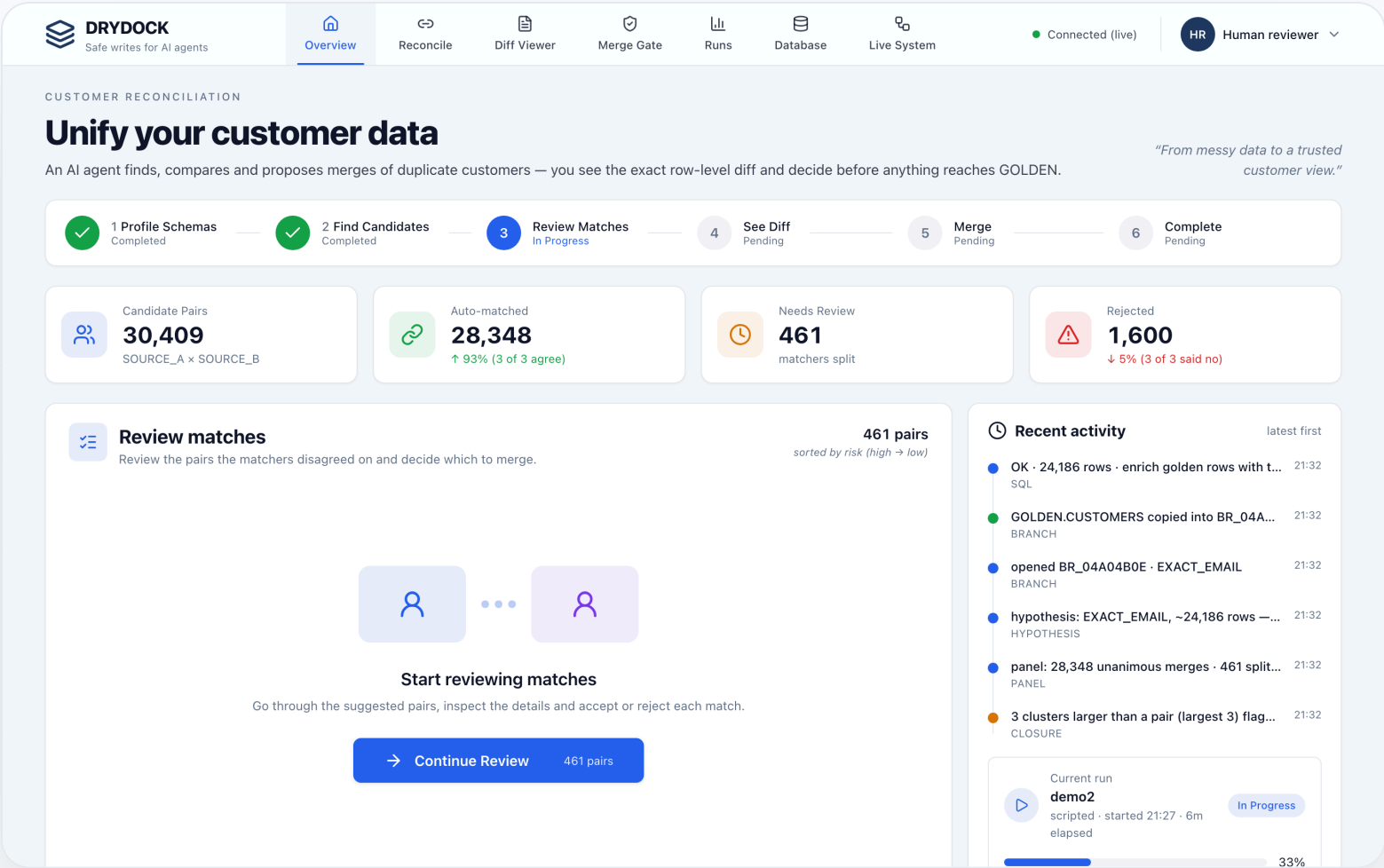
≈35 ms

throwing a branch away

Agents connect over MCP: Exasol's **official server for reads**, Drydock's for writes. Measured on Exasol Personal 2026.2 on a laptop.

Two messy systems, 69,000 records, 400 planted look-alikes.

- Three SQL matchers vote inside Exasol and settle **98.5%** of 30,409 candidate pairs
- Only the **461** they disagree on go to Gemini, as comparison summaries: never a name or an email
- Gemini's verdict is advice. Split rows still wait for a person



THE GATE AT WORK

The agent proposed ~28,000 merges. None reached production without passing the gate.

MERGED AUTOMATICALLY

4,413

new customers, within the tier's limits

HELD FOR A PERSON

229

split-vote rows across 3 match classes, kept out of the merge

DELETE GUARD

1.46%

deletes requested; the tier allows 1%, so a person decides

EXACT UNDO

Exact

unmerge restores the table; its fingerprint matches the original

A reviewer sees every changed row side by side, unticks what looks wrong, and merges the rest.
Rejections become precedents the adjudicator sees next time.

RESULTS ON THE LIVE INSTANCE

Measured against a ground truth the agent never sees.

27,980

of 27,981 proposed merges
correct (precision 0.99996)

99.93%

of true matches found (recall), F1
0.9996

600 / 600

internal duplicates found, none
wrong

38

live tests passing against Exasol

Every score is computed against a planted ground truth the agent has no access to, through a reviewer-only endpoint. The matcher, the gate and the database are each measured on their own.

Exasol's speed makes "branch production for every AI change" the default.

TODAY

- A safe write path next to Exasol's official, read-only MCP server
- Built only on native Exasol SQL, no extra services
- Every decision audited in the database

NEXT

- An engine-level zero-copy clone would make branches near-free at any scale
- An autonomous Gemini planner, already wired to the same MCP tools
- In-database adjudication once a model is deployed on the cluster



Governed changes. Observed impact.

Run it yourself: the README takes you from an empty Exasol to a live run.

github.com/paramvansh01/DryDock